

# 16: Product Review Classification using Function Calling

---

**Name: Y.VinodhKumar**

**RegNo: 192472359**

## Introduction

Product review classification is an important application of Large Language Models (LLMs) in e-commerce. Customers write reviews describing their experience with products. The objective is to automatically classify reviews as positive, negative, or neutral, detect fake reviews, summarize customer opinions, and identify important product features. Instead of relying only on text generation, function calling allows the LLM to securely interact with backend services and machine learning models. This improves accuracy, reduces hallucinations, and enables real-time integration with business applications.

Product review classification is an important application of Large Language Models (LLMs) in e-commerce. Customers write reviews describing their experience with products. The objective is to automatically classify reviews as positive, negative, or neutral, detect fake reviews, summarize customer opinions, and identify important product features. Instead of relying only on text generation, function calling allows the LLM to securely interact with backend services and machine learning models. This improves accuracy, reduces hallucinations, and enables real-time integration with business applications.

## Need for Function Calling

An LLM can understand the user's request but should not perform business operations by itself. Function calling enables the model to invoke backend APIs that classify reviews, calculate sentiment scores, detect spam, and retrieve product information. This separation improves reliability, maintainability, and security.

An LLM can understand the user's request but should not perform business operations by itself. Function calling enables the model to invoke backend APIs that classify reviews, calculate sentiment scores, detect spam, and retrieve product information. This separation improves reliability, maintainability, and security.

An LLM can understand the user's request but should not perform business operations by itself. Function calling enables the model to invoke backend APIs that classify reviews,

calculate sentiment scores, detect spam, and retrieve product information. This separation improves reliability, maintainability, and security.

## Architecture

Workflow:

Customer Review → LLM → Intent Detection → Function Calling Layer → Backend APIs → Classification Engine → Database → Response to User.

The LLM identifies the task and passes structured parameters to the correct backend function. The backend performs the actual computation and returns verified results.

Workflow:

Customer Review → LLM → Intent Detection → Function Calling Layer → Backend APIs → Classification Engine → Database → Response to User.

The LLM identifies the task and passes structured parameters to the correct backend function. The backend performs the actual computation and returns verified results.

Workflow:

Customer Review → LLM → Intent Detection → Function Calling Layer → Backend APIs → Classification Engine → Database → Response to User.

The LLM identifies the task and passes structured parameters to the correct backend function. The backend performs the actual computation and returns verified results.

## Functions

1. `classify_review(review_text, product_id)`

Purpose: Classify review as Positive, Negative, or Neutral.

2. `analyze_sentiment(review_text)`

Purpose: Generate sentiment score between -1 and +1.

3. `detect_spam_review(review_text, user_id)`

Purpose: Identify fake or promotional reviews.

4. `summarize_reviews(product_id)`

Purpose: Generate a summary of all customer reviews.

5. `extract_features(review_text)`

Purpose: Identify product features such as camera, battery, display, design, and price.

1. `classify_review(review_text, product_id)`

Purpose: Classify review as Positive, Negative, or Neutral.

2. `analyze_sentiment(review_text)`

Purpose: Generate sentiment score between -1 and +1.

3. `detect_spam_review(review_text, user_id)`

Purpose: Identify fake or promotional reviews.

4. `summarize_reviews(product_id)`

Purpose: Generate a summary of all customer reviews.

5. `extract_features(review_text)`

Purpose: Identify product features such as camera, battery, display, design, and price.

1. `classify_review(review_text, product_id)`

Purpose: Classify review as Positive, Negative, or Neutral.

2. `analyze_sentiment(review_text)`

Purpose: Generate sentiment score between -1 and +1.

3. `detect_spam_review(review_text, user_id)`

Purpose: Identify fake or promotional reviews.

4. `summarize_reviews(product_id)`

Purpose: Generate a summary of all customer reviews.

5. `extract_features(review_text)`

Purpose: Identify product features such as camera, battery, display, design, and price.

## Parameters

Typical parameters include `review_text`, `product_id`, `user_id`, `review_id`, `language`, `timestamp`, `confidence score`, and `category`. These parameters are validated before any function is executed.

Typical parameters include review\_text, product\_id, user\_id, review\_id, language, timestamp, confidence score, and category. These parameters are validated before any function is executed.

Typical parameters include review\_text, product\_id, user\_id, review\_id, language, timestamp, confidence score, and category. These parameters are validated before any function is executed.

## Workflow

Step 1: User submits a review.  
Step 2: LLM understands the intent.  
Step 3: Appropriate function is selected.  
Step 4: Parameters are extracted.  
Step 5: Backend API executes the function.  
Step 6: Result is returned to the LLM.  
Step 7: LLM presents a user-friendly response.

Step 1: User submits a review.  
Step 2: LLM understands the intent.  
Step 3: Appropriate function is selected.  
Step 4: Parameters are extracted.  
Step 5: Backend API executes the function.  
Step 6: Result is returned to the LLM.  
Step 7: LLM presents a user-friendly response.

Step 1: User submits a review.  
Step 2: LLM understands the intent.  
Step 3: Appropriate function is selected.  
Step 4: Parameters are extracted.  
Step 5: Backend API executes the function.  
Step 6: Result is returned to the LLM.  
Step 7: LLM presents a user-friendly response.

## Validation

The system checks that the review is not empty, the product ID exists, the review length is valid, the language is supported, duplicate reviews are avoided, and all parameters follow the expected format.

The system checks that the review is not empty, the product ID exists, the review length is valid, the language is supported, duplicate reviews are avoided, and all parameters follow the expected format.

The system checks that the review is not empty, the product ID exists, the review length is valid, the language is supported, duplicate reviews are avoided, and all parameters follow the expected format.

## **Security**

Use encrypted communication, authenticated APIs, role-based access control, audit logs, rate limiting, input validation, prompt injection protection, and privacy protection for customer information.

Use encrypted communication, authenticated APIs, role-based access control, audit logs, rate limiting, input validation, prompt injection protection, and privacy protection for customer information.

Use encrypted communication, authenticated APIs, role-based access control, audit logs, rate limiting, input validation, prompt injection protection, and privacy protection for customer information.

## **Advantages**

Function calling provides accurate results, integrates with existing e-commerce platforms, supports structured outputs, reduces hallucinations, improves scalability, enables real-time processing, and separates AI reasoning from business logic.

Function calling provides accurate results, integrates with existing e-commerce platforms, supports structured outputs, reduces hallucinations, improves scalability, enables real-time processing, and separates AI reasoning from business logic.

Function calling provides accurate results, integrates with existing e-commerce platforms, supports structured outputs, reduces hallucinations, improves scalability, enables real-time processing, and separates AI reasoning from business logic.

## **Limitations**

The system depends on backend APIs, requires proper authentication, may fail if APIs are unavailable, and needs continuous monitoring to prevent misuse or malicious prompts.

The system depends on backend APIs, requires proper authentication, may fail if APIs are unavailable, and needs continuous monitoring to prevent misuse or malicious prompts.

The system depends on backend APIs, requires proper authentication, may fail if APIs are unavailable, and needs continuous monitoring to prevent misuse or malicious prompts.

## **Conclusion**

Function calling significantly improves Product Review Classification by allowing the LLM to connect with trusted backend services. It produces accurate classifications, sentiment scores, summaries, and feature extraction while maintaining security, scalability, and enterprise reliability.

Function calling significantly improves Product Review Classification by allowing the LLM to connect with trusted backend services. It produces accurate classifications, sentiment scores, summaries, and feature extraction while maintaining security, scalability, and enterprise reliability.